

Erdős Problem #671

EP-671

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Given $a_i^n \in [-1, 1]$ for all $1 \leq i \leq n < \infty$ we define p_i^n as the unique polynomial of degree $n-1$ such that $p_i^n(a_i^n) = 1$ and $p_i^n(a_{i'}^n) = 0$ if $1 \leq i' \leq n$ with $i \neq i'$. We similarly define $\mathcal{L}^n f(x) = \sum_{1 \leq i \leq n} f(a_i^n) p_i^n(x)$, the unique polynomial of degree $n-1$ which agrees with f on a_i^n for $1 \leq i \leq n$ (that is, the sequence of Lagrange interpolation polynomials).